Emotion Detection in Social Media with Retrieval-Augmented Generation (RAG) Justification

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Motivation

- Social media: short, noisy, emotional and informal text
- Some models can predict emotions, but without clear explanation on how they do it

Research Objective

- ➢Emotion detection in social media

+

- ➢Providing interpretable evidence-based justifications for predictions

Dataset

DATASET NAME: GOEMOTIONS DATASET

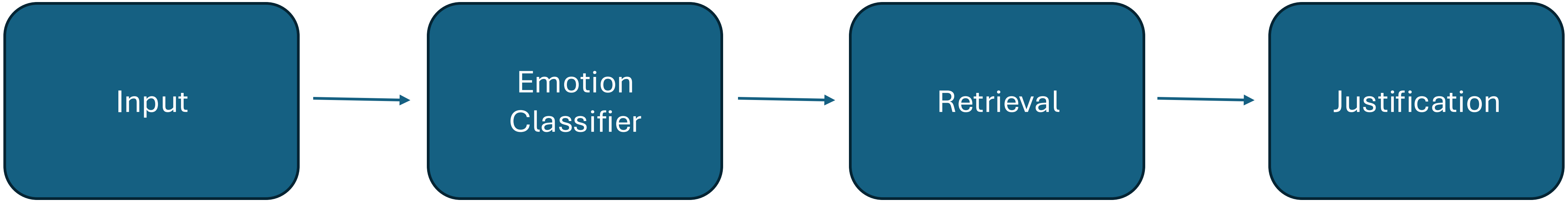
SOURCE: REDDIT

QUANTITY OF EMOTIONS: 27 EMOTIONS + NEUTRAL = 28

NATURE: MULTI-LABEL (ONE ELEMENT IN THE DATASET CAN HAVE 1+ LABELLED EMOTIONS)

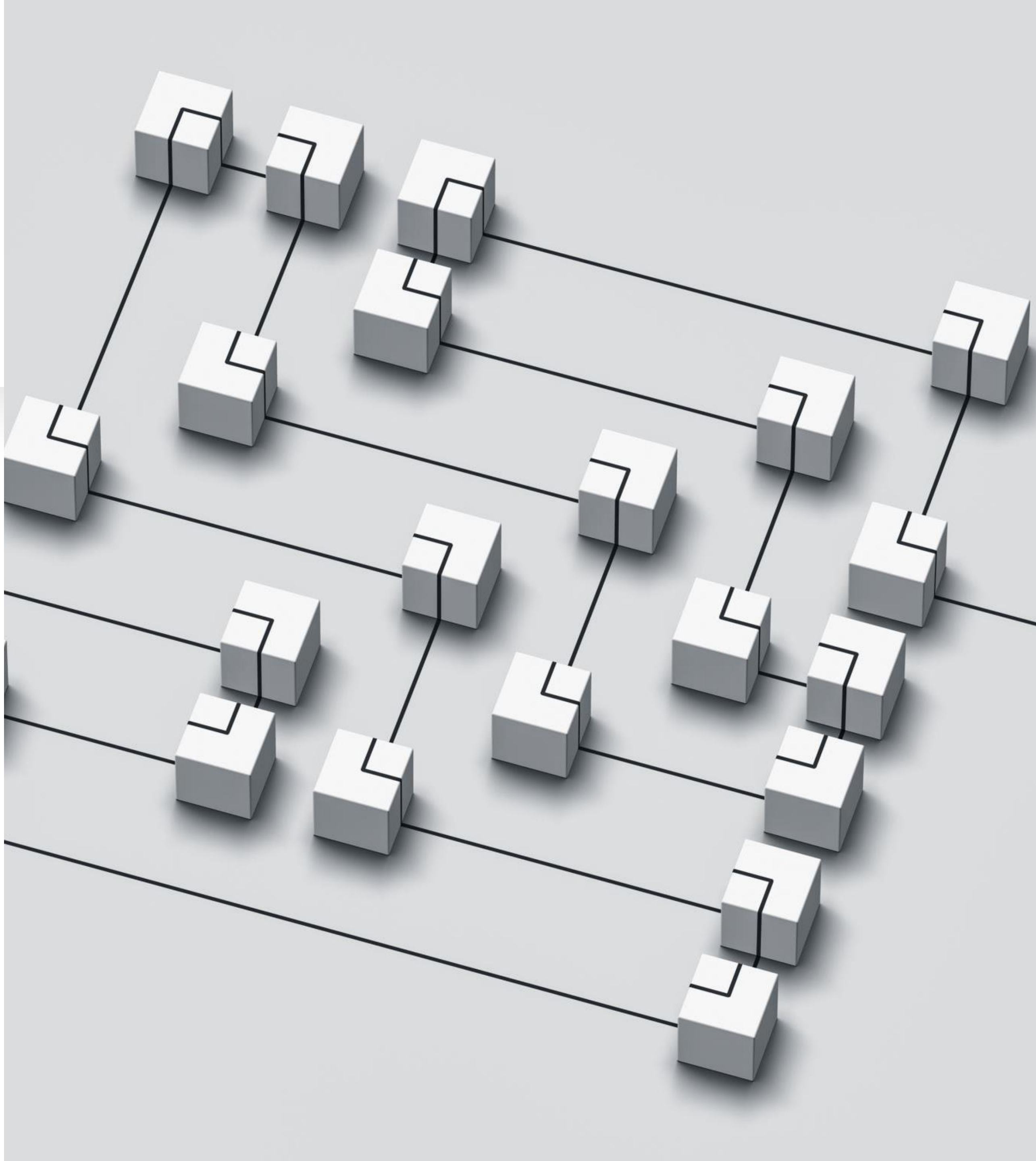
WHY THIS DATASET: FINE-GRAINED AND MULTI-LABEL, THEREFORE REALISTIC

System Pipeline



Emotion Classification Model

1. **RoBERTa:** high-performance transformer model that improves upon BERT by training on a significantly larger dataset
2. **Fine-tuning for multi-label classification**
3. **Sigmoid outputs**



Threshold Optimization

Baseline Global Threshold: = 0.5

Optimized Threshold: per-label thresholds

- A distinct decision threshold is set for each label

Benefit: Improved macro-F1 results

Retrieval-Augmented Justification

01

Retrieve semantically similar posts

02

Filter by shared emotions

03

Use examples as justification

Input: *“I feel empty and exhausted, nothing makes sense anymore.”*

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Predicted emotions: ['annoyance', 'disappointment', 'nervousness', 'sadness', 'neutral']

Retrieved similar posts (evidence):
(Ex1) labels=['annoyance', 'disappointment', 'sadness'] | Virtually no happiness anymore, exhausted with trying, meds not working properly and just tired of it
(Ex2) labels=['disappointment'] | Living is exhausting....
(Ex3) labels=['neutral'] | I've isolated myself to the extent that i only focus my energy on myself/work.as lonely as it sounds its been such a bre...
(Ex4) labels=['neutral'] | It's like how I feel after I eat McDonalds. Satisfied, disappointed, sad, full, tired....
(Ex5) labels=['sadness'] | I'm kind of struggling tonight. Bored, lonely... man...

Justification:
 [Ex4] It's like how I feel after I eat McDonalds. Satisfied, disappointed, sad, full, tired. [Ex5] I'm kind of struggling tonight. Bored, lonely... man
```

RAG Justification: Example

Experimental Results

- **Note**: the averaged F1-score documented for the GoEmotions dataset is **0.46**

Setting	DEV Micro-F1	DEV Macro-F1	TEST Micro-F1	TEST Macro-F1
Fixed threshold (0.5	0.510	0.469	0.501	0.453
Tuned per-label thresholds	0.598	0.540	0.598	0.521

Limitations

Retrieval quality is dependant on the dataset

Justification ≠ explanation

Conclusion

Task executed with the use of a **fine-tuned transformer-based classification model (RoBERTa)** and a **retrieval-augmented justification framework**

The system provides example-based justifications by retrieving social media posts with the most semantically similar meaning

Future Work

Justification quality can be increased

Alternative retrieval approaches can be more suited for the task in hand

Overall performance can always be optimized

Thank you for your attention!